

Homework # 6: Bayesian hypothesis testing and Bayesian prediction

Due March 21

1. A large shipment of parts is received, out of which five are tested for defects. The number of defects X has $\text{Binomial}(5, \theta)$ distribution. From the history of past shipments, it is known that θ has a $\text{Beta}(1, 9)$ prior distribution. Suppose that $X = 0$ is observed.
 - (a) Is there a significant evidence that the proportion of defective parts in the whole shipment exceeds 0.1? Conduct a Bayesian test comparing posterior probabilities of the null hypothesis $H_0 : \theta \leq 0.1$ and the alternative hypothesis $H_A : \theta > 0.1$.
 - (b) Revise your conclusion to the previous question taking into account that accepting a large shipment with more than 10% of defective parts is very costly, and thus, the Type II error results in a loss that is 3 times the loss due to the Type I error.
 - (c) Compute the Bayes factor for the test in (b,c).

Hint. Beta(1, β) distribution has pdf $f(x) = \beta(1 - x)^{\beta-1}$ and cdf $F(x) = 1 - (1 - x)^\beta$.

2. The probability ξ of the missile being destroyed by antimissile defense systems has the prior density

$$\pi(\xi) = 2(1 - \xi), \quad \xi \in (0, 1).$$

- (a) A priori, is there a significant evidence that the missile can be destroyed with probability less than 30%?
 - (b) Six missiles are launched, and none of them are shot down. Is there a significant evidence now that the probability of a missile being destroyed is less than 30%?
3. Six measurements $X_1, \dots, X_6$ from $\text{Normal}(\theta, 1)$ distribution are made. The first five are 3, 4, 5, 6, and 7, but X_6 got lost. Parameter θ is unknown; it has the $\text{Normal}(4, 1)$ prior distribution. Derive the predictive density of X_6 and a 90% predictive interval for it.
4. The probability θ to hit a certain target is deemed to have $\text{Beta}(5, 2)$ prior distribution. Among ten randomly selected shooters, five hit the target and five missed it. Now, you have two shots to make at the same target.
 - (a) Find the predictive distribution of Z , the number of times that you will hit the target.
 - (b) Your teammate proposes a bet. If you hit the target twice, she will pay you \$10. If you hit it only once, she will pay you \$2. If you miss it twice, you pay her \$15. Would you accept this bet, assuming a linear utility function between -\$20 and \$20?

5. The time it takes for a car to pass a certain segment of a highway has an Exponential distribution whose parameter depends on the number of open express lanes. There can be 2, 3, 4, or 5 open express lanes, with probabilities 0.1, 0.2, 0.4, and 0.3, respectively. If 2, 3, 4, or 5 express lanes are open, it takes on the average 60, 40, 35, and 25 minutes to pass this segment, respectively.

At the exit, three cars reported the passing times of 45, 42, and 48 minutes. Find the prediction density for the time it takes the next car to pass this segment.

6. * (Berger 4.63, [not required](#)). A missile can travel at either a high or a low trajectory. The missile's effectiveness decreases linearly with the distance by which it misses its target, up to a distance of 2 miles at which it is totally ineffective.

If a low trajectory is used, the missile is safe from antimissile fire. However, its accuracy is subject to the proportion of cloud cover θ . Indeed, the distance d , by which it misses its target is Uniformly distributed on $(0, \theta)$. For the target area, θ is known to have the probability density

$$\pi_1(\theta) = 6\theta(1 - \theta), \quad \theta \in (0, 1).$$

If a high trajectory is used, the missile will hit the target exactly, unless it is first destroyed by antimissile fire. From previous experience, the probability ξ of the missile being destroyed is thought to have the prior density

$$\pi_2(\xi) = 2(1 - \xi), \quad \xi \in (0, 1).$$

Two missiles are launched using a high trajectory, out of which none are shot down.

- (a) Give the loss incurred in having the missile miss by a distance d . Let 0 stand for a perfect hit and 1 for a total miss.
- (b) What is the optimal Bayes trajectory?

Hint for (b): what does this question have to do with Bayesian prediction? Well, to solve (b), you need to find the predictive density of d for the low trajectory and the predictive density of $Z = \{\text{indicator of a missile being shot}\}$ in the case of the high trajectory.